

CHAPTER 6

Offline-First TinyML–Federated Learning Framework for Privacy-Preserving and Energy- Efficient Smart Healthcare Systems

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ABSTRACT

Physiological data streams define continuous monitoring in smart healthcare environments. High-frequency signals are generated under strict latency and privacy constraints. Cloud-dependent architectures introduce transmission delays and expose sensitive patient information. Reliability is reduced in real-time decision scenarios. Edge devices impose strict limits on memory, energy, and computational capacity. Conventional deep learning models exceed these constraints, restricting deployment feasibility. Distributed inference enables localised processing under such conditions. TinyML is integrated with federated optimisation to support on-device intelligence. Model execution occurs on microcontroller-based wearable devices, enabling real-time interpretation of physiological signals such as cardiac activity and motion patterns. Training follows a decentralised update mechanism in which parameter aggregation replaces raw data transmission. Privacy is preserved while maintaining global model consistency. Offline-first operation ensures inference continuity through buffered synchronization under intermittent connectivity. System behaviour is governed by interaction between compression, decentralized learning, and adaptive execution. Quantization and pruning reduce memory footprint and computational demand. Federated updates regulate model generalization across heterogeneous data distributions. Runtime execution adapts to energy availability and processing load, ensuring bounded resource utilization. Empirical evaluation indicates improved detection reliability with reduced inference latency and lower energy consumption compared to cloud-centric approaches. Communication overhead is minimized through parameter-level synchronization. The framework enables scalable, privacy-preserving, and energy-efficient healthcare monitoring in resource-constrained and intermittently connected environments.

Keywords—TinyML, federated learning, edge intelligence, offline-first systems, privacy-preserving healthcare, microcontroller-based inference.

I. INTRODUCTION

Physiological monitoring environments generate high-frequency biomedical signals. Electrocardiograms, motion patterns, and oxygen saturation levels exhibit bounded